Experiment :- 2

Aim :- To convert an ER digram into relational tables using MySQL and apply constraints to maintain data accuracy and referential intergrity

Theory :- An ER diagram can be converted into relational tables. Each entity becomes a table, and its attributes become colums. Primary key uniquely identify each record while foreign keys connect related tables.

SQL Program :-

```
CRETE DATABASE CollegeDB;
USE CollegeDB;
CREATE TABLE Department (
        dept_id INT PRIMARY KEY,
        dept_name VARCHAR(50) Not NULL UNIQUE);

CREATE TABLE student (
        student_id INT PRIMARYKEY,
        name VARCHAR(50) Not NULL,
```

```
    dept_id INT,
    FOREIGN KEY (dept_id)
        REFERENCES Department (dept_id)
        ON DELETE SET NULL );

CREATE TABLE Course (
        Course_id INT PRIMARY KEY,
        Course_name VARCHAR(50) Not NULL UNIQUE,
        dept_id INT,
        FOREIGN KEY (dept_id)
            REFERENCES Department (dept_id)
            ON DELETE CASCADE );

INSERT INTO Department VALUES (1, 'CSE');
INSERT INTO Department VALUES (2, 'AI');

INSERT INTO student VALUES (101, 'Rahul', 1);
INSERT INTO student VALUES (102, 'Rohan', 2);
INSERT INTO Course VALUES (201, 'DBMS', 1);
INSERT INTO Course VALUES (202, 'Python', 2);

SELECT * FROM Department;
SELECT * FROM Course;
SELECT * FROM student;
```

Referential Integrity Violation :-

```
INSERT INTO student VALUES (103, 'RAVI', 99);
INSERT INTO student VALUES (101, 'Amit', 1);
```

Result:- MySQL Rejects both statements bcz the foreign key references a non-existing department and the primary key must be unique.

Important Constraints :-

* **PRIMARY KEY** :- Uniquely identifies each record.
* **FOREIGN KEY** :- Connects two related tables.
* **Not NULL** :- Does not allow empty values.
* **UNIQUE** :- Prevents duplicate values.
* **ON DELETE CASCADE** :- Delete related course records when their department is deleted.
* **ON DELETE SET NULL** :- Sets the students department ID to Null when that department is deleted

Conclusion :-

Thus, the ER daigram is Converted into relational tables using MY SQL, Different Constraints are applied to maintain data accuracy and referential Integrity.